

CAWSR: Carla-Autoware Scenario Runner

David Gasinski¹, Olek Osikowicz¹, Gwilym Rutherford¹, and
Donghwan Shin¹

¹ The University of Sheffield

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Summary

CAWSR facilitates the testing of the open-source autonomous driving system, Autoware, within CARLA, the state-of-the-art open-source driving simulator. Building on existing tools, this project introduces a research-oriented testing framework for the execution of complex driving scenarios, as well as supporting implementation of a wide range of verification strategies.

Statement of Need

The aim of this project is to address the technical challenges associated with the verification of state-of-the-art Autonomous Driving Systems (ADS) in virtual environments. Industry-grade systems, such as Autoware ([Kato et al., 2018](#)) or Apollo ([Baidu, 2017](#)), are inherently complex, requiring extensive configuration and are typically integrated into specialized simulators like AWSIM ([Tier4/AWSIM, 2025](#)) or the Apollo Game Engine Based Simulator ([Baidu, 2024](#)), respectively.

Conversely, CARLA ([Dosovitskiy et al., 2017](#)) has emerged as the de facto standard within the ADS research community. It holds a prominent position in the autonomous vehicle domain due to its rich ecosystem of open-source tooling. While alternatives such as BeamNG.tech ([Maul et al., 2021](#)) and AWSIM have recently gained traction, they are often designed for narrower applications. For instance, BeamNG.tech utilizes a soft-body physics engine to simulate high-fidelity vehicle dynamics, whereas AWSIM is tailored specifically for integrating a single AD system (Autoware). Despite these alternatives, CARLA remains the preferred choice for end-to-end testing in the research community, due to its support for a wide variety of driving systems, extensive testing benchmarks, and utility tools.

While a basic support for communication between Autoware and CARLA exists, it does not currently form a executable scenario-based testing framework, a leading approach for ADS verification.

This work bridges the gap by enabling the evaluation of Autoware in complex driving scenarios within CARLA. Adhering to the widely used CARLA platform rather than nascent alternatives, this work introduces a framework built directly upon the established CARLA ecosystem, facilitating direct comparison with the state-of-the-art driving agents evaluated on the CARLA Leaderboard.

The effective ADS verification requires the ability to systematically explore the operational design domain. To support this, the tool provides an extensible interface for algorithmic scenario generation and prioritization, facilitating a wide range of verification strategies based on common evaluation metrics such as the CARLA Leaderboard's driving score ([CARLA Team, 2024](#)).

Lastly, maintaining a consistent testing environment is essential, as simulators can introduce unintentional nondeterminism that leads to inconsistent evaluation results ([Osikowicz et al., 2025](#)).

41 Therefore, this framework seeks to minimize the introduction of additional nondeterminism
42 into the evaluation pipeline.

43 State of the Field

44 As previously established, CARLA offers a rich ecosystem of tools and documentation. Multiple
45 frameworks already exist for supporting popular open-source AD systems, such as *apollo-carla-*
46 *bridge* (Guardstrikelab, 2023) for Apollo and *autoware-carla-bridge* (Kaljavesi et al., 2024) for
47 Autoware. These tools solve a key issue; transforming sensor and control data from CARLA to
48 formats supported by the each system. However, native support for scenario execution engines is
49 non-existent, restraining their use for scenario-based testing and verification. These techniques
50 significantly reduce the effort required to validate autonomous driving systems, lowering the
51 technical barrier of entry compared to alternative approaches such as mileage-based testing,
52 which require high startup costs.

53 The current standard for scenario execution within CARLA is the CARLA Leaderboard and
54 its execution engine, Scenario Runner (SR) (CARLA, 2025). Common use cases of these
55 frameworks include testing end-to-end driving models, such as those based on VLM (Vision
56 Language Models) or Reinforcement Learning. No support is included for testing ROS based
57 AD systems, which focus on a modular approach to autonomous driving, rather than end-to-end.
58 PCLA (Tehrani et al., 2025), a framework for CARLA, addresses the topic of scenario-testing by
59 creating a clear deployment pipeline for autonomous agents / systems into CARLA, including
60 Autoware. However, their methodology focuses on the simplification of agent implementation
61 and abstraction of setup across various CARLA versions. While this allows for quick use and
62 evaluation without relying on external codebases (such as the CARLA Leaderboard), there is
63 a clear gap on deep intergration between the agent and simulator, limiting the execution of
64 complex route-based scenarios.

65 Tool Summary

66 CAWSR is a fully synchronous testing-framework directly integrating Scenario Runner. It is
67 built and deployed as a Docker container alongside Autoware and includes two main modes
68 of operation. *Algorithm* mode supports the execution of custom verification strategies and
69 algorithms implemented by the user. *Benchmark* mode includes functionality to execute
70 and evaluate a set of scenario definitions empirically. A fully synchronous pipeline has been
71 developed to ensure no new non-determinism is introduced, although some may exist directly
72 within the simulator itself (Osikowicz et al., 2025).

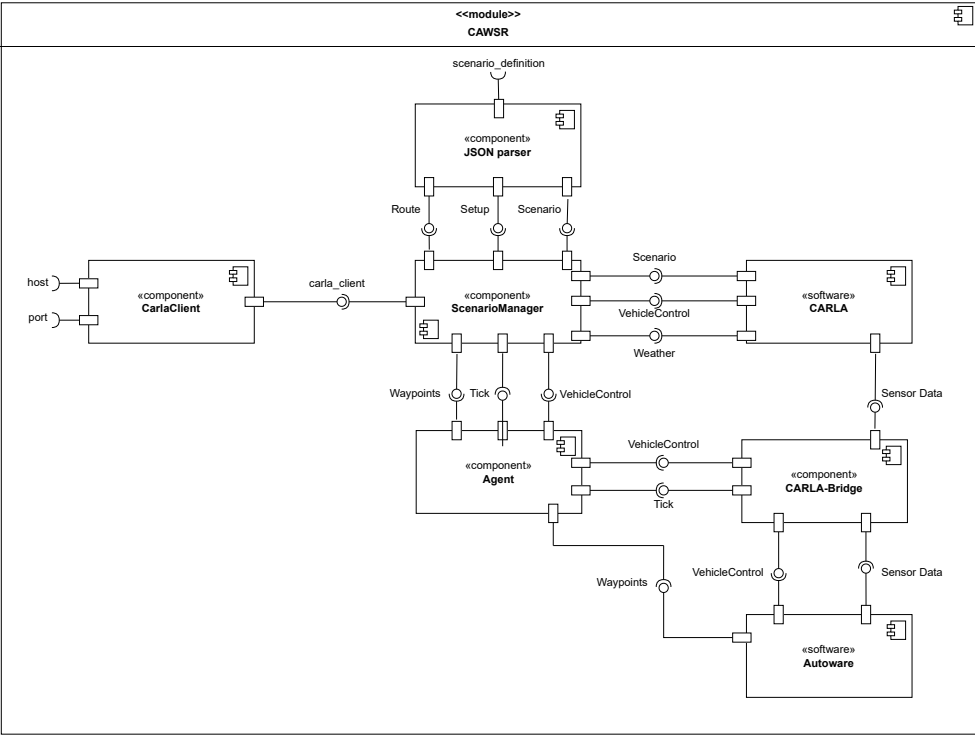


Figure 1: Internal component diagram of CAWSR.

To facilitate development, we introduce a new domain model for the definition of route-based scenarios within CARLA, alongside a JSON implementation. This model is based on the format introduced by Scenario Runner, facilitating support between both frameworks.

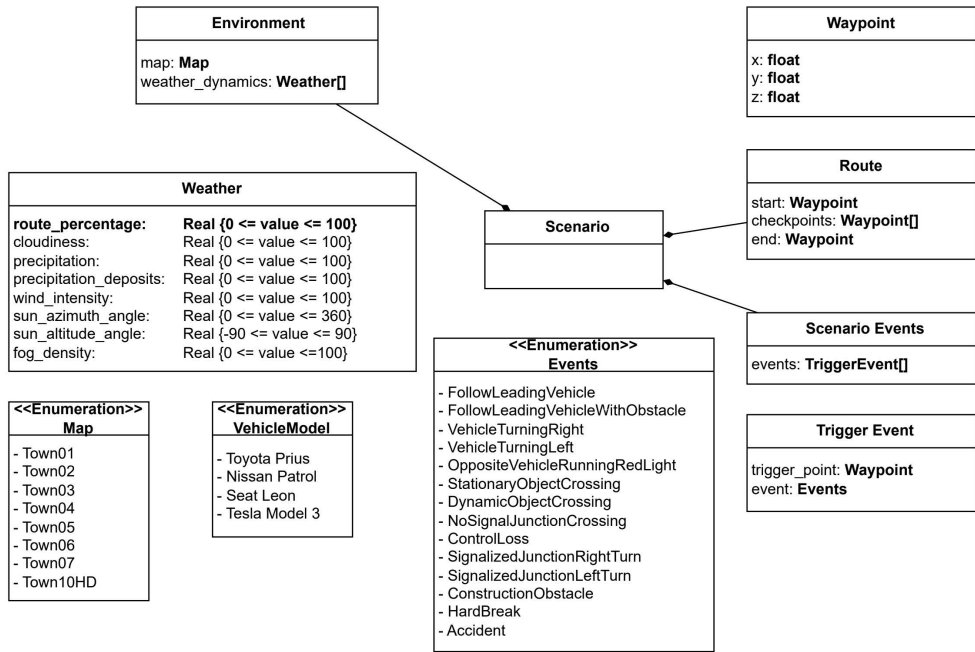


Figure 2: Scenario definition domain model.

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